

Setup Guide

Real-Time implementation of the NLMS Algorithm - EE310

Overview

This guide is for checkpoint 4, covering how to set up and use the physical implementation of the NLMS algorithm, divided into phases.

Hardware Setup

If using Windows 10/11, go to settings->system->sound->properties. Select the device's speaker and in the menu that opens, disable Audio enhancements. Do the same for the device's microphone.

Run tests to ensure the microphone and speaker are working.

Then, download all the necessary prerequisite code for Checkpoint 3.

Phase 0

Run the file called 'phase0.py'. If '[test1phase0.py](#)' is available as well, this may be an earlier iteration of the code used for testing.

A prompt will open for selecting the far end wave. Select any file from the "Mono audio" folder. Let the audio play until the end.

Then, play the output file generated in "Output/phase 0".

Phase 1, 2 & 3

Run the file called 'phase0.py'. If '[test1phase0.py](#)' is available as well, this may be an earlier iteration of the code used for testing.

A prompt will open for selecting the far end wave. Select any file from the "Mono audio" folder. Let the audio play until the end.

Then, play the output file generated in "Output/phase 0".